

FLAT ISLAND LIGHT, AK, USA

WMO: 994720

Lat: 59.332N

Lon: 151.995W

Elev: 59

StdP: 14.66

Time Zone: -9.00 (NAK)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.4	15.2	4.4	6.9	17.0	6.7	7.8	18.9	44.4	26.8	40.6	26.9	24.9	280	0.790

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.1	61.8	55.3	59.5	54.1	58.0	53.5	56.7	60.3	55.4	58.3	54.5	57.3	8.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
54.6	63.6	57.8	53.6	61.3	56.6	52.9	59.6	55.9	24.1	60.6	23.5	58.6	22.9	57.4	60.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 37.1	32.8	28.5	8.3	65.8	3.5	2.2	5.7	67.3	3.7	68.6	1.7	69.9	-0.9	71.4		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.7	32.6	32.3	32.6	37.4	44.8	49.4	53.5	54.4	50.7	43.6	35.4	33.3		
	DBStd	9.96	8.35	6.70	6.73	4.56	3.42	3.04	2.90	2.52	3.45	4.61	7.03	7.45		
	HDD50	3358	539	496	539	377	166	45	4	1	32	202	439	518		
	HDD65	8496	1003	916	1004	827	627	468	358	328	430	663	889	983		
	CDD50	335	0	0	0	0	4	27	111	138	52	3	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	12.2	15.6	15.3	14.1	10.4	8.2	7.6	7.5	8.0	11.2	13.9	17.6	17.1		
	PrecAvg	24.8	2.1	1.9	1.3	1.1	0.8	1.1	1.8	2.6	3.7	2.8	2.6	3.0		
(15) Precipitation	PrecMax	34.3	7.1	6.1	2.8	3.8	2.2	2.0	3.6	4.5	6.7	8.1	7.8	5.0		
	PrecMin	15.0	0.0	0.3	0.1	0.2	0.3	0.3	0.6	0.3	1.2	1.2	0.3	0.4		
	PrecStd	3.9	1.5	1.3	0.8	0.8	0.4	0.5	0.8	1.0	1.4	1.4	1.9	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.2	45.9	45.0	50.1	58.1	62.7	65.6	64.4	60.8	56.4	50.7	48.4		
		MCWB	44.0	41.7	41.3	44.4	48.4	53.0	56.3	57.7	55.2	50.8	45.9	44.3		
	2%	DB	46.4	42.8	43.4	46.1	53.4	58.0	61.5	61.8	57.9	52.6	48.0	45.6		
		MCWB	42.3	39.4	40.0	42.0	45.8	52.0	55.3	56.1	53.0	48.2	44.1	41.4		
	5%	DB	44.2	41.3	42.0	44.3	51.0	55.4	59.3	59.6	56.4	50.5	46.2	43.8		
		MCWB	41.0	38.4	39.2	40.9	45.7	50.9	54.2	54.6	52.1	46.7	42.8	40.5		
	10%	DB	42.2	40.0	40.6	42.8	49.5	53.4	57.5	58.2	55.1	49.3	44.3	42.0		
		MCWB	39.8	37.5	38.1	39.7	45.1	49.6	53.4	54.0	51.3	45.7	41.5	39.4		
	0.4%	WB	44.3	42.1	42.5	44.7	49.6	55.0	57.4	58.7	56.4	51.3	47.0	45.0		
		MCDB	48.1	44.3	44.0	48.9	55.0	59.1	62.8	62.4	59.9	55.0	49.1	47.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.8	40.5	41.2	42.9	48.0	53.0	56.1	57.2	53.9	49.2	45.0	42.5		
		MCDB	45.8	42.3	42.7	45.8	51.6	56.4	60.2	60.6	56.5	51.6	47.5	44.7		
	5%	WB	41.5	39.1	40.1	41.4	46.9	51.5	55.0	56.0	53.0	47.5	43.5	41.2		
		MCDB	43.9	40.8	41.9	43.7	50.1	55.1	58.5	58.8	55.4	49.4	46.0	43.2		
	10%	WB	40.1	38.0	38.8	40.2	45.9	50.5	54.1	55.1	52.1	46.2	42.0	39.7		
		MCDB	42.3	39.8	40.9	42.7	49.1	53.4	57.2	57.6	54.5	48.5	44.5	41.7		
	5% DB	MDBR	6.0	6.1	6.1	6.0	6.3	6.4	6.7	6.1	5.5	5.1	5.9	6.0		
		MCDBR	8.0	7.1	6.2	8.0	9.1	9.8	9.3	8.3	7.0	6.9	7.2	7.5		
		MCWBR	5.4	5.4	4.5	4.7	4.7	4.9	4.7	4.4	4.7	4.8	5.9	5.2		
		MCWBR	7.7	6.6	5.6	6.8	7.9	8.6	8.2	7.1	6.4	6.0	6.9	6.9		
(35) Mean Daily Temperature Range	5% WB	MCWBR	5.6	5.4	4.8	5.0	4.9	4.8	4.7	4.4	4.8	4.7	6.0	5.2		
	taub		0.265	0.310	0.319	0.360	0.378	0.366	0.369	0.362	0.340	0.322	0.284	0.272		
		taud	2.259	2.332	2.395	2.319	2.325	2.448	2.488	2.501	2.519	2.400	2.194	2.041		
(40) Clear-Sky Solar Irradiance	Ebn at Noon		176	225	261	267	270	276	272	266	253	217	157	112		
		Edh at Noon	13	21	26	34	36	32	30	28	23	19	14	11		
	RadAvg		104	276	693	1122	1539	1698	1503	1200	787	383	150	64		
(44) All-Sky Solar Radiation	RadStd		14	46	106	155	154	143	138	153	115	41	26	8		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		+1.51	+4.21	+3.92	N/A	N/A	N/A	-405	-517	N/A	N/A

Nomenclature: See separate page